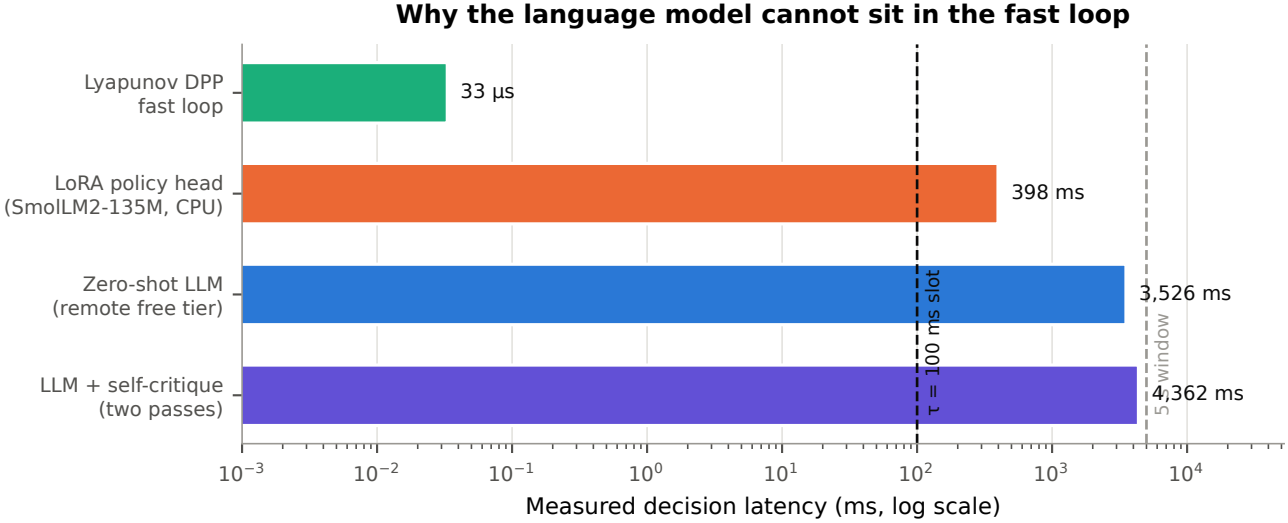
Why the language model cannot sit in the fast loop
Lyapunov DPP fast loop
33 µs
LoRA policy head (SmolLM2-135M, CPU)
398 ms
Zero-shot LLM (remote free tier)
3,526 ms
LLM + self-critique (two passes)
4,362 ms
τ = 100 ms slot
5 s window
10^−3
10^−2
10^−1
10^0
10^1
10^2
10^3
10^4
Measured decision latency (ms, log scale)